TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	88.3 ± 0.90
Correlation Stability	78.1 ± 1.14
Distribution Consistency	89.0 ± 0.77
Entropy Preservation	85.7 ± 0.46
Mean Z Score Drift	0.1 ± 0.02
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	99.8 ± 0.60
Feature Accuracy	93.0 ± 1.67
Novelty Score	0.9 ± 0.01

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Dataset Audit: breast_cancer_coimbra.csv

1. Dataset Portrait

Attribute	Value
Rows	116
Columns	10
Numeric features	10
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	57.302	58.913	+1.611	2.8%	OK
BMI	27.582	28.351	+0.769	2.8%	OK
Glucose	97.793	99.196	+1.403	1.4%	OK
Insulin	10.012	11.743	+1.731	17.3%	Warn
HOMA	2.695	3.044	+0.349	13.0%	Warn
Leptin	26.615	24.815	-1.800	6.8%	OK
Adiponectin	10.181	9.644	-0.537	5.3%	OK
Resistin	14.726	17.001	+2.275	15.4%	Warn
MCP.1	534.647	623.092	+88.445	16.5%	Warn
Classification	0.552	0.609	+0.057	10.3%	Warn

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	88.3 ± 0.90
Correlation Stability (*2)	78.1 ± 1.14
Distribution Consistency (*3)	89.0 ± 0.77
Entropy Preservation (*4)	85.7 ± 0.46
Mean Z Score Drift (*5)	0.1 ± 0.02
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00
Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	99.8 ± 0.60

Feature Accuracy (*10)	93.0 ± 1.67
Novelty Score (*11)	0.9 ± 0.01

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7291666666666667	0.4666666666666667	0.6875000000000001	-0.04166666666666664	
F1	0.7533203093203094	0.5295702632554691	0.7233824235264398	-0.029378857938695688	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.725	0.4749999999999999	0.6083333333333334	-0.11666666666666667	
F1	0.7359528619528619	0.5185801512169699	0.6271620996224577	-0.10879076233040427	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6416666666666667	0.5375000000000001	0.6	-0.04166666666666666	
F1	0.5557236965932618	0.5393282969753559	0.4486705564966345	-0.10705314009661837	*

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7208333333333334	0.5125000000000001	0.6416666666666667	-0.07916666666666668	
F1	0.7258341658341658	0.5329173252595413	0.6590596678268488	-0.06677449800731709	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6375	0.5583333333333333	0.6333333333333333	-0.004166666666666691	
F1	0.5764264088314431	0.6156609403668227	0.7025460076750398	0.12611959884359675	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7083333333333334	0.5333333333333333	0.625	-0.08333333333333334	*
F1	0.7179811943844927	0.6027625628578975	0.6659466337127627	-0.05203456067172999	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7458333333333333	0.4416666666666667	0.675	-0.07083333333333337	
F1	0.7685622114156597	0.4700516295853623	0.7124952138401297	-0.05606699757553006	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.4625	0.5125	0.4749999999999999	0.01249999999999994	
F1	0.5361601301923884	0.5775608831587092	0.5525050055617353	0.01634487536934699	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.4958333333333333	0.475	0.4916666666666666	-0.00416666666666668	
F1	0.6547289527475286	0.5494874211003242	0.6135632041514395	-0.04116574859608915	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7458333333333333	0.4874999999999993	0.6250000000000001	-0.12083333333333332	*
F1	0.754839950419102	0.4993909145905514	0.6500292863336341	-0.10481066408546783	*

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	48.0	23.5	70.0	2.707	0.467408667	8.8071	9.7024	7.99585	417.114	0.0
2	83.0	20.69049454	92.0	3.115	0.706897333	8.8438	5.429285	4.06405	468.786	0.0
3	82.0	23.12467037	91.0	4.498	1.009651067	17.9393	22.43204	9.27715	554.697	0.0
4	68.0	21.36752137	77.0	3.226	0.612724933	9.8827	7.16956	12.766	928.22	0.0
5	86.0	21.11111111	92.0	3.549	0.8053864	6.6994	4.81924	10.57635	773.92	0.0
6	49.0	22.85445769	92.0	3.226	0.732086933	6.8317	13.67975	10.3176	530.41	0.0
7	89.0	22.7	77.0	4.69	0.890787333	6.964	5.589865	12.9361	1256.083	0.0
8	76.0	23.8	118.0	6.47	1.883201333	4.311	13.25132	5.1042	280.694	0.0
9	73.0	22.0	97.0	3.35	0.801543333	4.47	10.358725	6.28445	136.855	0.0
10	75.0	23.0	83.0	4.952	1.013839467	17.127	11.57899	7.0913	318.302	0.0
11	34.0	21.47	78.0	3.469	0.6674356	14.57	13.11	6.92	354.6	0.0
12	29.0	23.01	82.0	5.663	1.145436133	35.59	26.72	4.58	174.8	0.0
13	25.0	22.86	82.0	4.09	0.827270667	20.45	23.67	5.14	313.73	0.0
14	24.0	18.67	88.0	6.107	1.33	8.88	36.06	6.85	632.22	0.0
15	38.0	23.34	75.0	5.782	1.06967	15.26	17.95	9.35	165.02	0.0
16	44.0	20.76	86.0	7.553	1.6	14.09	20.32	7.64	63.61	0.0
17	47.0	22.03	84.0	2.869	0.59	26.65	38.04	3.32	191.72	0.0
18	61.0	32.03895937	85.0	18.077	3.790144333	30.7729	7.780255	13.68392	444.395	0.0
19	64.0	34.5297228	95.0	4.427	1.037393667	21.2117	5.46262	6.70188	252.449	0.0
20	32.0	36.51263743	87.0	14.026	3.0099796	49.3727	5.1	17.10223	588.46	0.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	42.0	36.51263743	83.0	12.305	0.612724933	50.53	8.42996	31.6904	632.22	0.0
2	66.0	31.25	118.0	3.33	0.801543333	31.1233	1.65602	16.43706	392.46	1.0
3	45.0	22.85445769	90.0	51.814	0.755688	8.8438	18.55	17.10223	358.624	1.0
4	40.0	27.1	95.0	3.881	2.534931667	45.272	20.03	10.57635	206.802	0.0
5	49.0	32.5	101.0	21.699	2.2485936	25.7816	5.4861	12.06534	923.886	0.0
6	77.0	24.24242424	80.0	16.635	0.706897333	8.88	9.73138	16.1	1078.359	1.0
7	64.0	22.22222222	98.0	5.7	1.37788	12.1905	4.783985	13.91245	395.976	1.0
8	57.0	34.83814777	95.0	12.548	2.940414667	33.1612	2.36495	9.9542	655.834	1.0
9	68.0	21.08281329	102.0	6.2	1.55992	9.6994	8.574655	13.74244	448.799	1.0
10	42.0	21.35991456	93.0	2.999	0.6879706	19.0826	8.462915	17.37615	321.919	1.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	74.0	23.5	70.0	3.226	13.22733227	10.16	10.22231	5.62592	378.996	1.0
2	28.0	36.7901662	102.0	9.208	0.921719333	35.891	6.966895	31.6904	1698.44	1.0
3	68.0	22.86	131.0	5.782	1.009651067	7.6476	22.43204	9.35	225.88	1.0
4	49.0	27.68877813	92.0	8.15	0.507936	28.562	21.57	4.82	232.006	1.0
5	48.0	18.67	85.0	6.524	1.55992	19.0653	20.03	5.768	269.487	0.0
6	65.0	27.88761707	85.0	4.713	1.449709333	89.27	23.67	31.6904	335.393	1.0
7	50.0	38.57875854	106.0	6.703	1.752611067	46.6401	4.667645	11.78388	887.16	0.0
8	57.0	34.83814777	95.0	12.548	2.940414667	33.1612	2.36495	9.9542	655.834	1.0
9	38.0	22.4996371	95.0	5.261	1.232827667	8.438	4.77192	15.73606	199.055	1.0
10	81.0	31.64036818	100.0	9.669	2.38502	38.8066	10.636525	29.5583	426.175	1.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	85.0	29.2184076	131.0	4.364	25.05034187	9.8	17.95	11.78796	695.754	1.0
2	59.0	30.48315806	134.0	14.649	1.846290133	50.53	4.81924	24.3701	99.45	1.0
3	45.0	22.86	93.0	3.226	25.05034187	9.62	4.138025	28.0323	199.055	1.0
4	38.0	34.83814777	102.0	9.245	1.1174	18.16	9.349775	23.1177	244.75	1.0
5	54.0	32.03895937	87.0	4.713	0.742368	21.2117	7.780255	3.32	773.92	0.0
6	57.0	26.6	139.0	14.649	5.271762467	10.16	16.67	24.3701	280.694	1.0
7	43.0	31.25	103.0	4.328	1.099600533	25.7816	12.71896	38.6531	775.322	1.0
8	24.0	18.67	88.0	6.107	1.33	8.88	36.06	6.85	632.22	0.0
9	71.0	25.51020408	112.0	10.395	2.871792	19.0653	5.4861	42.7447	799.898	1.0
10	49.0	21.36752137	78.0	2.64	0.507936	6.3339	3.886145	22.94254	737.672	1.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	72.0	29.4	87.0	24.887	0.6538048	13.08	3.71009	4.06405	256.001	0.0
2	66.0	21.08281329	83.0	3.549	1.4066068	17.9973	22.43204	5.06	359.232	1.0
3	45.0	20.83	103.0	15.533	0.605507467	50.6094	7.16956	4.6638	806.724	1.0
4	71.0	31.25	101.0	2.869	2.3498848	8.8071	10.73174	9.9542	960.246	1.0
5	47.0	30.91557669	82.0	6.524	1.432235467	17.9973	8.574655	9.6135	313.73	1.0
6	85.0	27.63605442	94.0	2.707	2.241625267	26.8081	8.2863	8.2049	209.749	0.0
7	75.0	30.48	152.0	7.01	2.628282667	50.53	10.06	11.73	99.45	1.0
8	60.0	26.34929208	103.0	5.138	1.305394533	24.2998	2.19428	20.2535	378.996	0.0
9	77.0	35.58792924	76.0	3.881	0.727558133	21.7863	8.12555	17.2615	618.272	0.0
10	72.0	23.62	105.0	4.42	1.14478	21.78	17.86	4.82	195.94	1.0

B.5 Audit Trace: Fold 5 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	43.0	28.57667585	106.0	4.53	6.777364	4.311	4.267105	16.48508	209.19	1.0
2	86.0	27.88761707	112.0	10.395	8.225983067	13.08	21.42	10.57635	764.667	0.0
3	69.0	24.21875	77.0	14.026	0.972138	17.9973	4.138025	5.1042	218.28	1.0
4	47.0	30.83653053	90.0	6.76	1.883201333	90.28	2.78491	23.03408	713.239	1.0
5	71.0	32.03895937	78.0	4.376	7.836205333	28.562	5.589865	5.768	572.783	1.0
6	71.0	27.3	106.0	5.819	1.1006464	4.47	5.1	10.69548	1102.11	1.0
7	77.0	35.58792924	76.0	3.881	0.727558133	21.7863	8.12555	17.2615	618.272	0.0
8	48.0	28.125	90.0	2.54	0.56388	15.5325	10.22231	16.11032	1698.44	1.0
9	58.0	29.15451895	139.0	16.582	5.685415067	22.8884	10.26266	13.97399	923.886	1.0
10	68.0	21.36752137	77.0	3.226	0.612724933	9.8827	7.16956	12.766	928.22	0.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	34.0	34.5297228	89.0	5.138	2.098344	31.0385	2.19428	10.15726	573.63	1.0
2	76.0	32.46191136	89.0	6.47	3.495982	17.87	8.13	17.55503	358.624	0.0
3	45.0	23.62	93.0	9.245	1.089637667	14.3224	4.230105	13.74244	353.568	1.0
4	45.0	23.12467037	199.0	6.59	2.05239	14.57	9.7024	10.96	864.968	0.0
5	72.0	23.14049587	93.0	4.462	2.5101466	11.2406	13.494865	5.31	358.624	1.0
6	47.0	36.21227888	105.0	4.56	6.4834952	7.85	6.160995	31.6904	199.055	1.0
7	74.0	28.65013774	88.0	3.012	0.6538048	31.1233	7.65222	18.35574	572.401	1.0
8	65.0	32.05	97.0	5.73	1.370998	61.48	22.54	10.33	314.05	1.0
9	76.0	27.2	94.0	14.07	3.262364	35.891	9.34663	8.4156	377.227	0.0
10	65.0	29.666548	85.0	14.649	3.071407	26.5166	7.28287	19.46324	1698.44	1.0

B.7 Audit Trace: Fold 7 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	45.0	27.88761707	131.0	3.012	1.383997333	14.09	5.589865	8.04375	293.123	1.0
2	45.0	35.09270153	82.0	2.82	0.617890133	24.2998	8.12555	5.1042	473.859	1.0
3	89.0	28.44444444	131.0	5.197	0.617890133	18.16	8.412175	23.03306	244.75	1.0
4	75.0	26.6	75.0	4.09	9.736007333	14.3224	5.46262	14.91922	638.261	0.0
5	67.0	28.67262608	86.0	5.646	0.827270667	17.022	16.67	11.50005	632.22	1.0
6	60.0	20.83	97.0	5.646	5.091856133	46.076	3.335665	4.19	198.4	1.0
7	35.0	30.27681661	84.0	4.376	0.9067072	39.2134	9.048185	16.43706	733.797	0.0
8	69.0	32.5	93.0	5.43	1.245642	15.145	11.78796	11.78796	270.142	0.0
9	58.0	29.15451895	139.0	16.582	5.685415067	22.8884	10.26266	13.97399	923.886	1.0
10	49.0	29.77777778	70.0	8.396	1.449709333	51.3387	10.73174	20.76801	602.486	1.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	50.0	23.01	84.0	4.713	3.869788067	6.633	8.574655	17.10223	209.19	1.0
2	43.0	30.8012487	88.0	2.82	2.3498848	14.3224	4.77192	4.19	960.246	0.0
3	45.0	26.6	84.0	5.819	1.449709333	4.47	17.95	3.29175	136.855	1.0
4	49.0	27.7	82.0	6.59	4.668906667	27.1841	5.429285	14.91922	621.273	1.0
5	49.0	26.34929208	134.0	6.817	1.432235467	21.2117	6.209635	10.34455	209.19	1.0
6	89.0	26.66666667	118.0	9.245	3.775036	45.272	18.55	11.50005	488.829	0.0
7	45.0	37.03560819	83.0	6.76	1.383997333	39.9802	4.617125	8.70448	586.173	0.0
8	61.0	32.03895937	85.0	18.077	3.790144333	30.7729	7.780255	13.68392	444.395	0.0
9	46.0	22.21	86.0	36.94	7.836205333	10.16	9.76	5.68	312.0	1.0
10	85.0	27.68877813	196.0	51.814	25.05034187	70.8824	7.901685	55.2153	1078.359	1.0

B.9 Audit Trace: Fold 9 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	65.0	22.22222222	94.0	8.808	1.55992	45.6196	3.19209	26.0136	200.976	0.0
2	34.0	26.5625	104.0	4.462	1.099600533	14.7485	9.7024	10.96	220.66	1.0
3	47.0	29.60676726	94.0	41.611	2.940414667	17.127	3.886145	10.37518	887.16	0.0
4	66.0	28.125	90.0	4.376	2.38502	33.1612	8.01	11.78388	269.487	0.0
5	45.0	31.23140988	105.0	3.33	2.940414667	7.6476	11.57899	9.1539	530.41	0.0
6	85.0	23.12467037	83.0	2.869	0.732193	8.88	8.13	17.2615	586.173	0.0
7	78.0	25.3	60.0	3.508	0.519184	6.633	10.567295	4.6638	209.749	0.0
8	52.0	30.8012487	87.0	30.212	6.4834952	29.2739	6.26854	24.24591	764.667	1.0
9	45.0	20.26	92.0	3.44	0.780650667	7.65	16.67	7.84	193.87	1.0
10	59.0	28.67262608	77.0	3.188	0.605507467	17.022	16.44048	31.6904	910.489	1.0

B.10 Audit Trace: Fold 10 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	72.0	36.05	98.0	4.53	1.449709333	53.4997	4.267105	24.6033	314.05	1.0
2	35.0	35.58792924	78.0	10.491	0.612724933	6.964	17.95	19.46324	220.66	0.0
3	38.0	31.23140988	92.0	4.172	1.304866667	35.59	9.76	5.68	733.797	0.0
4	61.0	28.125	95.0	7.01	3.495982	14.9037	6.966895	6.92	209.19	0.0
5	29.0	27.7	77.0	4.58	2.628282667	90.28	8.300955	7.32	1698.44	1.0
6	69.0	19.13265306	138.0	5.75	3.775036	38.8066	11.78796	5.57055	1698.44	0.0
7	64.0	34.5297228	95.0	4.427	1.037393667	21.2117	5.46262	6.70188	252.449	0.0
8	32.0	36.51263743	87.0	14.026	3.0099796	49.3727	5.1	17.10223	588.46	0.0
9	43.0	31.25	103.0	4.328	1.099600533	25.7816	12.71896	38.6531	775.322	1.0
10	57.0	34.83814777	95.0	12.548	2.940414667	33.1612	2.36495	9.9542	655.834	1.0